

# Engineering Change Order ECO-2025-014

Product: ACT-FP-5500 'Liberty' Composite Frac Plug | Revision D → E

|                       |                                                                     |
|-----------------------|---------------------------------------------------------------------|
| ECO Number / Date     | ECO-2025-014 / 2025-11-20                                           |
| Originator            | R. Callahan                                                         |
| Affected Product      | ACT-FP-5500 'Liberty' Composite Frac Plug                           |
| Revision Change       | D to E                                                              |
| Change Classification | Class 1 (performance-affecting)                                     |
| Effectivity           | All shipments after 2025-11-20. First article serial FP55-25-18211. |

## 1. Reason for Change

Field data from 3 pads (612 plugs) showed 1.8% incidence of incomplete set in 23 lb/ft casing at measured depths beyond 19,000 ft MD. Analysis attributed marginal setting force (42,000 lbf with 6 screws) against higher element friction at elevated mud weights.

## 2. Description of Change

Increase setting shear screws from six (6) to eight (8) x 1/4-28 UNF brass C36000 (part ACT-FP-5500-10), raising nominal setting force from 42,000 lbf to 55,000 lbf. Mandrel screw bosses already provisioned with 8 holes (2 previously plugged). Update setting tool guidance: ACT-ST-10 output remains adequate (60,000 lbf max).

## 3. Affected Documents

| Document              | From Rev | To Rev | Change Summary                            |
|-----------------------|----------|--------|-------------------------------------------|
| BOM-FP-5500           | D        | E      | Item 11 qty 6 -> 8; note added            |
| DS-FP-5500            | D        | E      | Setting force spec updated                |
| MAN-FP-5500           | D        | E      | Sec 1 note; Sec 3 screw verification step |
| Dwg D-FP-5500-10      | C        | D      | Qty and pattern                           |
| ST-10 run sheet RS-19 | -        | -      | Charge selection table                    |

## 4. Validation / Qualification Evidence

Setting force verified on 10-plug qualification lot: mean shear-out 55,400 lbf, sigma 900 lbf (Test report TR-25-201). Full API 19V V3 requalification of Rev E completed 2025-11-14: 10,000 psi / 300 degF, all acceptance criteria met.

## 5. Disposition of Existing Stock and WIP

Rev D finished goods (stock: 1,240 plugs): USE AS IS for depths < 17,500 ft MD and casing <= 20 lb/ft, per Engineering Bulletin EB-25-09. Rev D plugs are NOT authorized for 23 lb/ft casing below 17,500 ft. WIP: rework to Rev E (add 2 screws).

## 6. Implementation Checklist

| Task                           | Owner              | Status   |
|--------------------------------|--------------------|----------|
| Update drawings and BOM in PLM | Design Engineering | COMPLETE |
| Update datasheet and manual    | Document Control   | COMPLETE |

| Task                                 | Owner                     | Status    |
|--------------------------------------|---------------------------|-----------|
| Update router / work instructions    | Manufacturing Engineering | COMPLETE  |
| Train assembly & QC personnel        | Quality                   | COMPLETE  |
| Notify affected customers / bulletin | Sales Engineering         | COMPLETE  |
| First-article inspection on next lot | Quality                   | SCHEDULED |

## 7. Approvals

| Role          | Name              | Signature               | Date       |
|---------------|-------------------|-------------------------|------------|
| Engineering   | J. Martinez, P.E. | /signed electronically/ | 2025-11-18 |
| Quality       | S. Nguyen         | /signed electronically/ | 2025-11-19 |
| Manufacturing | T. Okafor         | /signed electronically/ | 2025-11-19 |
| Doc Control   | L. Hargrove       | /signed electronically/ | 2025-11-20 |